

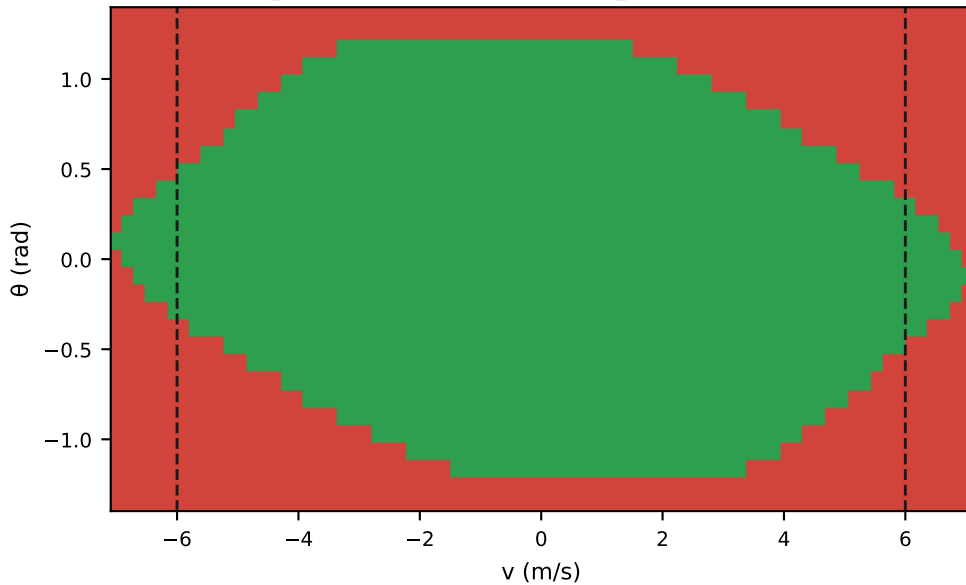
# v2.2 model layer — corrected bounds, avoid-only baseline, and reach-avoid on the corrected model

2026-07-29 · all quantities from hash-locked release artifacts · companion to CHANGES\_TO\_THE\_MODEL\_LAYER.md §§1-13 · page 1/2

MODEL LAYER (what changed under the math):  $a_{\text{tip}}$  6.998922 → 6.924595 (the old value used  $h_{\text{cm}} + \text{wheel\_radius}$  — the SPRUNG body's axle height mixed with the WHOLE robot's ground offset; corrected height 0.188559, not 0.196756, and the 7.276 mm lateral CoM offset is now carried).  $a_{\text{tip\_laden}}$  6.32652 → 5.326611, read from the laden grid's own metadata. ODD velocity  $\pm 4 \rightarrow \pm 6$  by operator ruling (monotone in g: everything previously certified stays certified). Speed governor now evaluates the SIGNED state — the  $|v|, |\dot{\psi}|$  fold assumed a symmetry the plant does not have and erred both ways (316 permissive / 510 conservative at reserve 0.35, 40k states/ $\mu$ ).

## Avoid-only GRID certificate ( $\mu=1.0$ , $\theta=\psi=0$ )

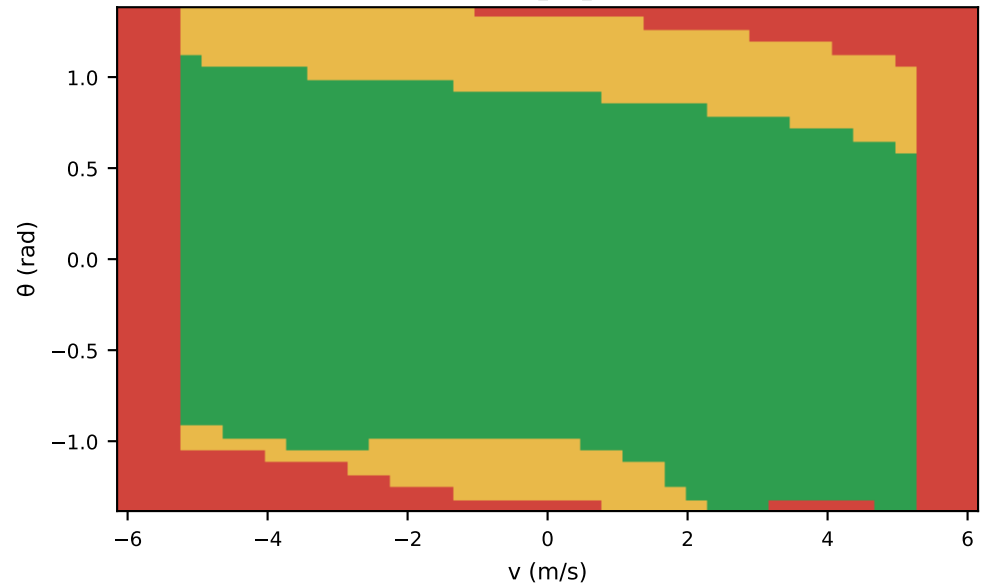
reduced 4-state  $f_{\text{cert}}$  · 1,161,508 nodes · corrected  $a_{\text{tip}} = 6.9246$  · window  $\pm 7$  (dashed =  $\pm 6$  ODD)



reached (61.7%)    limbo — safe, no reach (17.3%)    failed (21.0%)    certified safe (left)

## REACH-AVOID rollouts on the corrected v2.2 model

MuJoCo contact physics · 300k steps, scratch\_raw\_sat ·  $\mu=0.8$ , 3 s horizon, 41×41



READ BEFORE COMPARING WITH THE ORIGINAL 3-PANEL FIGURE— the domains differ. Your panels ran  $v \in [-0.5, 1.5]$ ; this is the full  $\pm 6$  ODD (6× wider), and the checkpoint is 300k steps and NOT converged: 17.3% limbo is the signature of an undertrained reach policy (your converged arms collapse limbo to 0%). Authorized for EVALUATION ONLY— the critic still carries the inherited max-entropy term inside the DRABE backup (§11; your ISAACS code already excludes it: base\_block.py:406). Measured on this run:  $\alpha$  auto-tuned to  $1.4e-4$ ,  $\beta < 0$  in 99.3% of states — contamination ran conservative here, by sign luck, not by guarantee.

LEFT PANEL LIMITS (§10):  $\min_d$  over disturbance-box corners is optimistic (3/1,161,508 states left the set under denser sampling; an exact fix was attempted and REVERTED — the bad-mask discontinuity defeats the affine-per-cell argument). Early-stopping residual is UNBOUNDED (iters/decade degrade 63 → 3,153; no geometric tail exists — your non-contraction result, measured). Node-exhaustive is not a continuum proof. Roll constraint is quasi-static on a roll-free model; the validation plant drops the 7.276 mm lateral offset (§13).

PLANTS DIFFER BY DESIGN: the left panel is the reduced 4-state  $f_{\text{cert}}$  (opt6 kernel — the same kernel the iLQR linearizes); the right panel is full MuJoCo contact physics. Green-left = "the reduced model certifies it"; green-right = "one MuJoCo rollout reached the target". VERIFICATION: 28 release gates, 9/9 bounds, frame hygiene both trees, 15/15 + 124 controller, 18/18 SSB, 152 toolchain — all green on quiescent trees. Preconditions satisfied BY CONSTRUCTION (§12.1): finite control set,  $\forall g = g$ . Warm-starting VI destroys both.

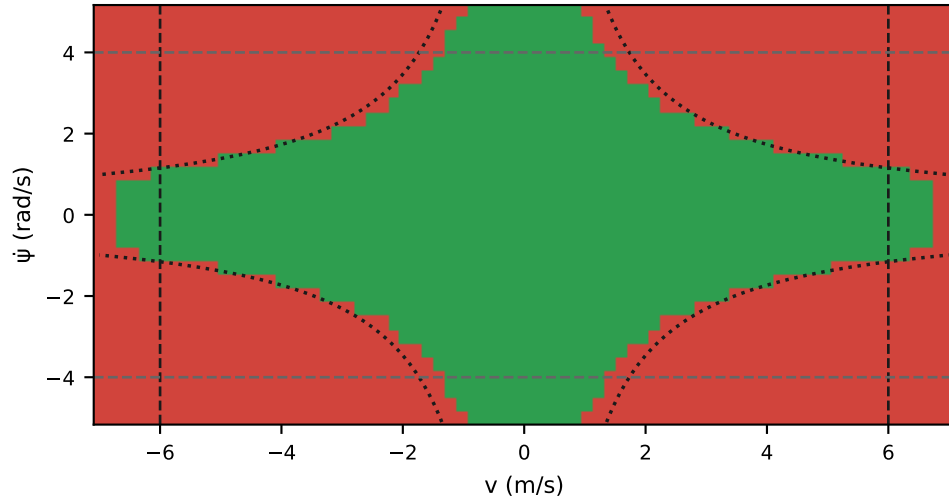
# Forward velocity vs yaw velocity — certified turn authority and the roll hyperbola

avoid-only capability grids, corrected  $a_{\text{tip}} \cdot \theta = \theta = 0$  slices · page 2/2

TURN AUTHORITY — the  $(v, \psi)$  plane at  $\theta = \theta = 0$  answers "how hard can it turn at speed". The dotted curves are the ANALYTIC roll hyperbola  $|\psi| = a_{\text{tip}}/|v|$  from the corrected bound; the certified edge should hug it wherever roll (not friction) binds. Checked exhaustively: the certified yaw edge never exceeds the hyperbola at any of 606 speed-slices across both grids — at  $\mu \geq 0.6$  it sits exactly on the last grid node beneath it. At low  $\mu$ , friction binds first at low speed and roll takes over by  $v \approx 3$  m/s. Laden: the 5.327 bound costs turn authority (−15% at  $v = 2$ ,  $\mu=0.3$ ), NOT straight-line speed —  $a_{\text{tip}}$  bounds the PRODUCT  $v \cdot \psi$ , so tightening it squeezes turning. Any laden derate belongs on the yaw channel, not the speed channel. Dashed lines:  $\pm 6$  ODD ( $v$ ),  $\pm 4$  ODD ( $\psi$ ).

## UNLADEN $\mu=0.3$ (ice)

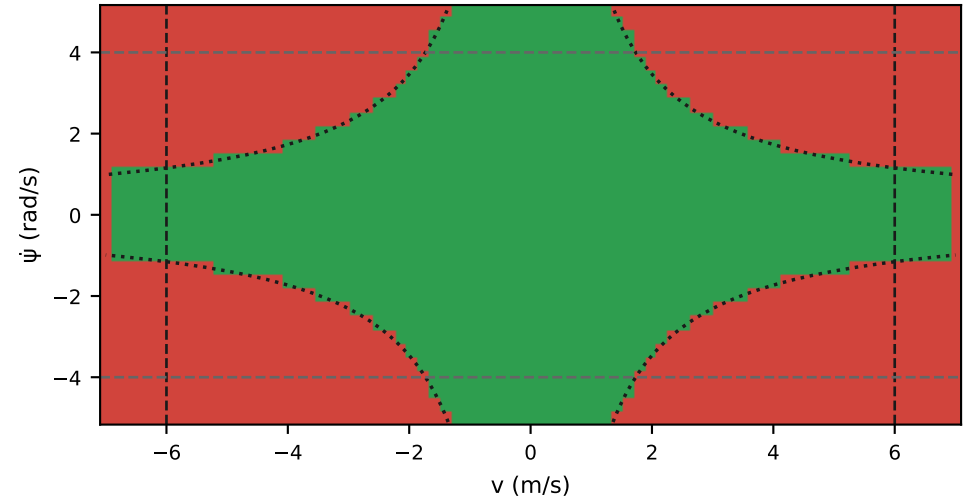
$a_{\text{tip}} = 6.925$  · friction binds at low  $v$ , roll by  $v \approx 3$



max  $|\psi|$  at  $v = 2$ : 3.00 · 4: 1.67 · 6: 1.00 rad/s

## UNLADEN $\mu=1.0$ (grip)

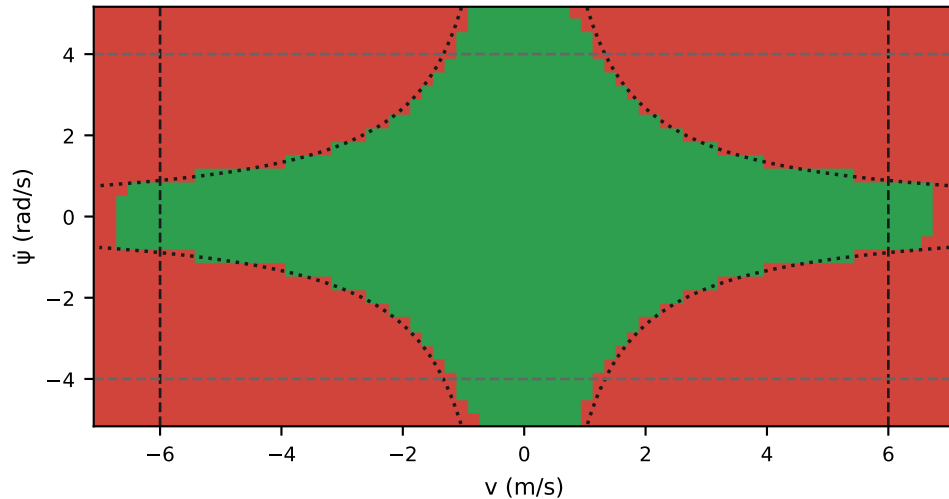
$a_{\text{tip}} = 6.925$  · edge on the hyperbola at all speeds



max  $|\psi|$  at  $v = 2$ : 3.33 · 4: 1.67 · 6: 1.00 rad/s

## LADEN $\mu=0.3$ (ice, $dh=0.30$ $dm=0.20$ )

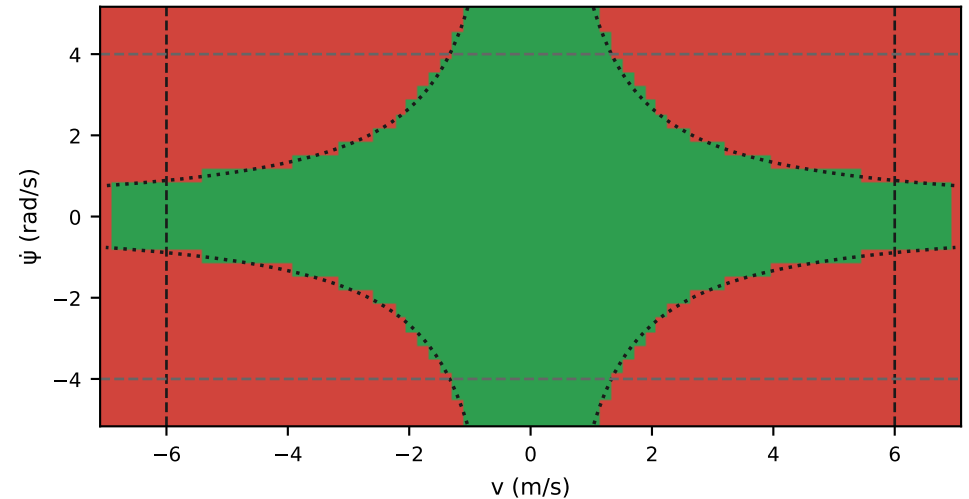
$a_{\text{tip}} = 5.327$  · tightest configuration in the release



max  $|\psi|$  at  $v = 2$ : 2.33 · 4: 1.00 · 6: 0.67 rad/s

## LADEN $\mu=1.0$ (grip)

$a_{\text{tip}} = 5.327$  · roll-bound everywhere



max  $|\psi|$  at  $v = 2$ : 2.67 · 4: 1.00 · 6: 0.67 rad/s